DataCollectionandPreprocessingPhase

Date	22April2026
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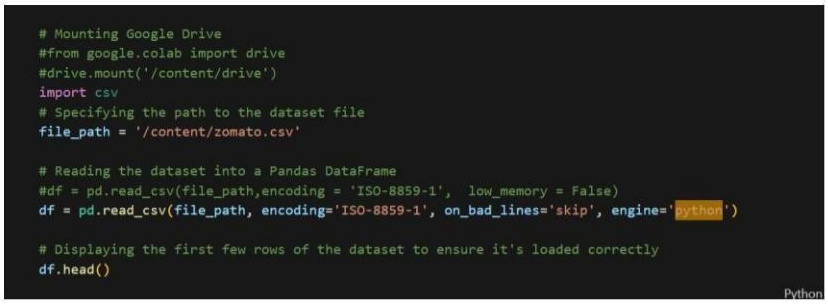
ProjectTitle	RestaurantRecommendation System
MaximumMarks	6 Marks

DataPreprocessing

The images will be preprocessed by resizing, normalizing, augmenting, denoising, adjusting contrast,detectingedges,convertingcolorspace,cropping,batchnormalizing,andwhitening data. These steps will enhance data quality, promote model generalization, and improve convergenceduringneuralnetworktraining,ensuringrobustandefficientperformanceacross various computer vision tasks.

Section	**Description**
DataOverview	Thedatasetcontainsrestaurantinformationfrom Zomato, includingname,reviews,ratings,cuisines,cost,andmore.The data is cleaned, deduplicated, and preprocessed for building a content-based recommendation system.
Resizing	*Notapplicablefortextdata.*

Normalization	Ratingsarenormalizedtoa1-5scaleusingMinMaxScaler.Text is lowercased and punctuation is removed.
DataAugmentation	Notapplicablefortext data.
Denoising	Textiscleanedbyremovingnewlinecharactersand punctuation.
EdgeDetection	Notapplicablefortext data.

ColorSpaceConversion	Notapplicablefortext data.
ImageCropping	Notapplicablefortext data.
Batch Normalization	Notapplicablefortext data.
DataPreprocessingCodeScreenshots	
LoadingData	
Resizing	*Not applicable*
Normalization	
DataAugmentation	*Not applicable*
Denoising	
EdgeDetection	*Not applicable*

```python
# Mounting Google Drive
#from google.colab import drive
#drive.mount('/content/drive')
import csv
# Specifying the path to the dataset file
file_path = '/content/zomato.csv'

# Reading the dataset into a Pandas DataFrame
#df = pd.read_csv(file_path,encoding = 'ISO-8859-1',  low_memory = False)
df = pd.read_csv(file_path, encoding='ISO-8859-1', on_bad_lines='skip', engine='python')

# Displaying the first few rows of the dataset to ensure it's loaded correctly
df.head()
```

```python
# Computing Mean Rating
restaurants = list(df['name'].unique())
df['Mean Rating'] = 0
for i in range(len(restaurants)):
  df['Mean Rating'][df['name'] == restaurants[i]] = df['rate'][df['name'] == restaurants[i]].mean()
#Scaling the mean rating values
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler (feature_range = (1,5))
df[['Mean Rating']] = scaler.fit_transform(df[['Mean Rating']]).round(2)
```

```python
## Lower Casing
df["reviews_list"] = df["reviews_list"].str.lower()
## Removal of Puctuations
import string
PUNCT_TO_REMOVE = string.punctuation
def remove_punctuation(text):
  """custom function to remove the punctuation"""
  return text.translate(str.maketrans('', '', PUNCT_TO_REMOVE))
df["reviews_list"] = df["reviews_list"].apply(lambda text: remove_punctuation (text))
```

ColorSpaceConversion	*Not applicable*
ImageCropping	*Not applicable*
Batch Normalization	*Not applicable*